

MOLTRIX

Professional Cloud, AI & Platform Engineering Roadmap

VERSION 3.1 • JULY 2026

Academic alignment • industry credentials • Moltrix projects • Canadian employment strategy

Executive judgment: Approved direction—conditional on execution discipline. Certifications validate knowledge; projects, testing, troubleshooting, communication and professional evidence create employability.

Strategic purpose

Build a durable Finance + Cloud + AI identity by converting George Brown coursework, prior Applied AI learning and 19 years of finance experience into current credentials, secure hands-on systems, GitHub evidence, professor-reviewed demonstrations and realistic Canadian employment pathways.

Existing strength	Capability being built	Target professional identity
Finance/accounting, controls, reporting, budgeting, audit and leadership	Microsoft 365, endpoint administration, Azure, automation, platform engineering, AI operations and later AWS	AI-enabled enterprise cloud professional with finance, governance and transformation depth

Professional outcome

Anwar can design, build, secure, automate, monitor, troubleshoot and improve cloud systems; deploy and govern AI responsibly; translate technical performance into business value; and demonstrate every major claim through Moltrix evidence.

Why This Roadmap?

The sequence is designed around employability, learning transfer and credible proof—not around collecting the largest number of certificates.

Why Microsoft first. George Brown coursework, completed desktop foundations and the Ontario enterprise market make Microsoft 365, Intune, Entra and Azure the fastest route to coherent practical evidence.

Why MD-102 first. Endpoint administration creates an attainable entry point while strengthening identity, policy, compliance, security and troubleshooting skills used later in Azure.

Why Azure before AWS. Depth in one cloud produces better judgment. Azure also connects directly to Microsoft 365, Entra, Intune and the current academic program.

Why AZ-104 before AZ-305. Architecture judgment should follow operational experience with identity, networking, compute, storage, monitoring, backup, recovery and cost.

Why platform engineering. Employers need repeatable systems, not portal-only demonstrations. Git, scripting, Infrastructure as Code, containers and CI/CD turn configuration into an engineering capability.

Why AI throughout. AI remains active in every phase. Advanced AI engineering follows cloud, data, security and automation foundations so AI systems can be deployed and governed responsibly.

Why every credential produces a project. A Moltrix project converts exam objectives into architecture, implementation, testing, troubleshooting and communication evidence.

Why Moltrix. Moltrix is the proof system: a consistent business context that demonstrates progression from endpoint operations to cloud platforms, AI operations and enterprise transformation.

Career principle: The objective is to become a professional who can design, build, secure, automate, govern and improve AI-enabled cloud solutions—not someone who merely lists certifications.

Roadmap Governance

Design rules

- Every major credential must produce a Moltrix project and public-safe GitHub evidence.
- At least 40% of technical study time must be hands-on configuration, testing, troubleshooting or automation.
- Only one major certification is prepared at full intensity at a time; adjacent foundations may run lightly in parallel.
- No advanced exam is booked until its associated project reaches demonstrable portfolio quality.
- AI continuity receives 5–7 protected hours weekly, even during infrastructure-heavy phases.
- Certification names, exam codes, objectives, retirement dates and prerequisites are checked on official pages before study activation and booking.
- Quarterly reviews may reorder future phases based on Ontario job demand, academic workload, cost, credential changes and demonstrated progress.

Capability and credential labels

Label	Meaning	Decision rule
REQUIRED CAPABILITY	Skill must be demonstrated	Project evidence required; exam may not be
ACTIVE CREDENTIAL	Current primary examination	One at full intensity
SUPPORTING FOUNDATION	Short adjacent learning	Must not delay active project
CONDITIONAL CREDENTIAL	Potentially valuable later	Activate only after prerequisites and market review
FUTURE SPECIALIZATION	Advanced or role-specific	Pursue only when target roles justify it

Employment targets

First-entry roles: Microsoft 365 Support • Endpoint/Intune Administrator • Cloud Support • Junior Azure Administrator • Infrastructure Support • Identity & Access • Information Systems Analyst

Finance-powered bridge roles: Finance Systems • ERP/Business Systems • Technology Risk • IT Governance • Cloud FinOps • Finance Transformation • AI/Cloud Governance

Current priority: MD-102 is the active major credential and Moltrix Secure Endpoint Management is the active evidence project. AB-900 may remain a short supporting foundation but must not delay endpoint implementation.

Master Roadmap — Foundations to Azure Operations

Phase 0 — Continuous Technical Foundation

Target: Continuous

Transition logic: This foundation supports every later endpoint, cloud, automation and AI decision.

Purpose: Prevent shallow cloud knowledge and weak troubleshooting.

Capabilities: Networking, DNS, identity, Windows, Linux, virtualization, permissions, Git basics and PowerShell/Bash/Python automation.

Moltrix proof: Moltrix OS & Network Lab plus troubleshooting journal.

Exit gate: Can explain and test identity, DNS, routing, permissions, operating systems and virtualization basics.

Phase 1 — Endpoint Administration & Security

Target: Jul–Sep 2026

Transition logic: Endpoint operations apply the foundation to real users, devices, policies and security controls.

Purpose: Create the fastest credible bridge from desktop foundations into enterprise cloud administration.

Capabilities: MD-102 objectives: deployment strategy, Intune enrollment, identity, configuration, applications, compliance, endpoint security and support.

Moltrix proof: Moltrix Secure Endpoint Management.

Exit gate: Core workflows implemented and tested; failures documented; public-safe GitHub evidence complete; exam-ready.

Phase 2 — Microsoft 365 AI Administration

Target: After official scope is verified

Transition logic: Secure endpoint and identity knowledge is the prerequisite for governing AI-enabled Microsoft 365 services.

Purpose: Extend Microsoft 365 administration toward Copilot, agents, security, governance and responsible adoption.

Capabilities: AB-900 knowledge; AB-650 remains planned/conditional until Microsoft publishes and stabilizes its official credential page, objectives and booking path.

Moltrix proof: Moltrix AI Services Administration & Governance Lab.

Exit gate: Official scope verified; licensing and lab path feasible; identity, governance and operational scenarios tested.

Master Roadmap — Azure & Employability

Phase 3 — Azure Fundamentals

Target: Fall 2026

Transition logic: Microsoft 365 administration now expands into general Azure infrastructure and governance.

Purpose: Build a correct model of cloud responsibility, services, governance, security and cost.

Capabilities: AZ-900 knowledge: cloud concepts, Azure architecture and services, management and governance.

Moltrix proof: Moltrix Azure Foundation: resource map, identity model, governance baseline and cost estimate.

Exit gate: Can explain IaaS/PaaS/SaaS, regions, identity, governance, security and cost using a Moltrix scenario.

Phase 4 — Azure Administration

Target: Fall–Winter 2026

Transition logic: Azure fundamentals become operational skill through deployment, control, monitoring and recovery.

Purpose: Build the central technical capability for Azure support and junior administration roles.

Capabilities: AZ-104: identity, governance, storage, compute, networking, monitoring, backup, recovery and cost control.

Moltrix proof: Moltrix Secure Azure Environment.

Exit gate: Access and network boundaries, monitoring, backup, restore, failure handling and cost controls tested; exam-ready.

Phase 5 — Employability Workstream

Target: Begin now; intensify before Jan 2027

Transition logic: Employability runs alongside technical phases so evidence is reviewed and improved while it is being built.

Purpose: Convert progress into Canadian opportunity before the entire roadmap is complete.

Capabilities: GitHub portfolio, résumé, LinkedIn, professor checkpoints, STAR stories, technical demonstrations, networking and targeted applications.

Moltrix proof: Professor-ready demonstrations and employer-facing portfolio reviews.

Exit gate: Two strong demonstrations; professor feedback recorded; résumé and LinkedIn aligned; targeted applications underway.

Phase 6 — Platform Engineering Foundation

Platform Engineering Foundation means learning to build repeatable, secure and observable delivery systems—not merely using cloud portals.

Sequence	Required capability	Demonstrated outcome
1	Git & GitHub	Repositories, commits, branches, pull requests, issues and documentation
2	PowerShell & Bash	Repeatable administration, validation and troubleshooting tasks
3	Bicep	Azure-native Infrastructure as Code with parameters, modules and controlled deployments
4	Terraform	Reusable infrastructure, state awareness, plans, modules and multi-cloud patterns
5	Docker	Secure container images, configuration, networking, storage and registries
6	GitHub Actions	CI/CD workflows with linting, testing, approval, deployment and rollback
7	Security & governance	Least privilege, secret handling, policy checks, supply-chain awareness and cost controls
8	Monitoring	Logs, metrics, alerts, deployment health and failure diagnosis

Moltrix practical proof

Moltrix Automated Cloud Deployment: provision an Azure environment with Bicep or Terraform, package a small application with Docker, and deploy through GitHub Actions with secure secrets, tests, monitoring and a documented rollback.

Required exit gate

- Infrastructure can be reproduced from a clean environment.
- Pipeline success, controlled failure and rollback are demonstrated.
- Repository scanning confirms no secrets or sensitive identifiers are exposed.
- Architecture, security decisions, test results, troubleshooting and cost are documented.

Credential decisions

Terraform Associate (004): optional milestone after at least two Infrastructure-as-Code deployments; project competence comes first.

Kubernetes/CKA: future specialization—not a Phase 6 commitment. Reassess only after Linux, containers, networking, IaC, CI/CD and monitoring are demonstrated and target roles justify Kubernetes administration.

Anti-overload rule: Do not add Kubernetes to Phase 6 merely to make the roadmap look advanced. A smaller reproducible platform project is stronger evidence than unfinished tool exposure.

Master Roadmap — AI, Architecture & Multi-Cloud

Phase 7 — Production AI Engineering

Target: 2027; after secure cloud foundations

Transition logic: The platform engineering delivery system now supports secure, repeatable AI deployment.

Purpose: Reconnect prior Applied AI capability with governed enterprise deployment.

Capabilities: Current Azure AI/Foundry path; RAG, agents, evaluation, monitoring, identity, secrets, cost, safety and human approval.

Moltrix proof: Moltrix AI Operations Assistant deployed behind authentication.

Exit gate: Grounding, safety, escalation, cost, monitoring and human approval evaluated with recorded tests.

Phase 8 — Azure Architecture

Target: After operational Azure experience

Transition logic: Administration and platform automation provide the evidence needed to make defensible architecture decisions.

Purpose: Build architecture judgment from real deployment, failure, recovery and cost experience.

Capabilities: AZ-305 path: identity, governance, monitoring, data, business continuity and infrastructure design.

Moltrix proof: Moltrix Enterprise Azure Architecture and architecture decision records.

Exit gate: Can explain security, availability, recovery, performance, operations and cost trade-offs.

Phase 9 — AWS Architecture

Target: 2027+; after Azure depth and AWS labs

Transition logic: One-cloud depth creates a reference model for comparing a second cloud intelligently.

Purpose: Create credible multi-cloud breadth without weakening Azure depth.

Capabilities: AWS Solutions Architect—Associate (SAA-C03) capability: IAM, VPC, compute, storage, databases, resilience, monitoring, recovery and cost.

Moltrix proof: Moltrix AWS Architecture plus Azure/AWS decision comparison.

Exit gate: Core services are deployed and tested before examination; provider trade-offs are explained using one business scenario.

Phase 10 — Business AI, Consulting & Transformation

Target: Ongoing destination

Transition logic: Technical breadth becomes enterprise value when it improves decisions, controls or operations.

Purpose: Combine technical capability with finance leadership, governance, ROI and organizational change.

Capabilities: Business analysis, responsible AI, FinOps, controls, adoption and measurable value.

Moltrix proof: Moltrix business AI case in forecasting, finance intelligence, quality or inventory.

Exit gate: System is deployed, secured, monitored and governed; decision impact, risk, adoption and ROI are measured.

Credential & Capability Policy

Item	Status	Activation rule	Required proof
MD-102	ACTIVE	Current primary credential	Secure endpoint project
AB-900	SUPPORTING	Short foundation; must not delay MD-102	M365 AI administration knowledge map
AB-650	CONDITIONAL	Official scope, release, licensing and lab feasibility verified	AI services governance lab
AZ-900	FOUNDATION	Complete as knowledge or credential based on schedule	Azure foundation map
AZ-104	REQUIRED NEXT	After endpoint foundation and Azure basics	Secure Azure environment
Terraform Associate (004)	OPTIONAL	After two IaC deployments	Reusable IaC repositories
CKA	FUTURE SPECIALIZATION	Only if target roles justify it	Container and platform foundations first
AZ-305	DEFER	After Azure operational experience	Architecture decision records
AWS SAA-C03	DEFER	After AWS labs and Azure depth	AWS deployment and cloud comparison

Locked decisions

- **KEEP:** MD-102, AB-900 foundation, AZ-900 foundation, AZ-104 and the project-first evidence model.
- **CONDITIONAL:** AB-650 after official release and verification; Terraform Associate after project competence.
- **DEFER:** AZ-305 until Azure operations; AWS SAA-C03 until AWS labs; CKA until role demand and prerequisites justify it.
- **REMOVE FROM FORWARD PLAN:** MS-102 as a new target if retirement timing makes completion impractical.
- **KNOWLEDGE ONLY:** AWS Cloud Practitioner unless voucher-supported or required by a specific opportunity.

Quarterly validation questions

1. Is the credential still current, available and relevant to target Ontario roles?
2. Has the associated Moltrix project reached demonstrable quality?
3. Does the next phase strengthen the target role—or merely add another badge?
4. Are cost, licensing, academic workload and lab access sustainable?
5. What professor, peer or employer feedback should change the next quarter?

Execution System & Evidence Standard

The ten-part Moltrix evidence chain

#	Evidence layer	Completion test
1	Business problem	A realistic organizational need, users, constraints and success measures.
2	Official skill mapping	Current exam or capability objectives mapped to deliverables.
3	Architecture	Components, identities, trust boundaries, data flows and decisions.
4	Implementation	Sanitized configuration, scripts, Infrastructure as Code and reproducible steps.
5	Security & governance	Least privilege, secrets, policy, privacy, risk and responsible AI.
6	Testing	Expected behavior, failed behavior, recovery and acceptance criteria.
7	Troubleshooting	Symptoms, hypotheses, evidence, remediation and lessons.
8	Operations & cost	Monitoring, support, recovery, capacity and financial impact.
9	AI extension	Responsible AI opportunity where it materially improves the solution.
10	Communication	Technical report, executive summary, demonstration and interview story.

Universal advancement gate

A phase advances only when Anwar can: EXPLAIN the purpose and trade-offs • BUILD the solution • SECURE the boundaries • TEST success and failure • TROUBLESHOOT evidence • DOCUMENT decisions • DEMONSTRATE the outcome.

Public evidence safety

GitHub evidence must remove passwords, tokens, recovery codes, tenant or subscription identifiers, personal information, device serial numbers and confidential organizational data. Sanitization is part of the professional standard, not an afterthought.

Operating Cadence & Job-Readiness Gates

Cadence	Required action	Output
Daily	Focused learning plus hands-on work	Lab note or meaningful commit when evidence exists
Weekly	Review objectives, project evidence, blockers and AI continuity	Progress log and next-week priorities
Biweekly	Complete one demonstrable project increment	Tested feature, diagram, script or decision record
Monthly	Portfolio and communication review	README update, sanitized evidence and one interview story
Quarterly	Roadmap revalidation	Credential, market, cost, workload and sequence decision

Professor and employer checkpoints

- Checkpoint 1 — Present the business problem, architecture and first tested control.
- Checkpoint 2 — Demonstrate failure handling, troubleshooting and security reasoning.
- Checkpoint 3 — Present the completed case study and request specific improvement feedback.
- Employment gate — Two strong demonstrations, aligned résumé and LinkedIn, six prepared STAR stories, targeted applications and an active networking list.

Consulting review for every proposed step

Required submission: My proposed next step is _____. It supports Phase _____. The official skill is _____. I will build _____. My GitHub evidence will be _____. It will take _____. The prerequisite is _____. The cost/licensing risk is _____.

Decision labels: **ALIGNED • ALIGNED—LATER • OPTIONAL • DISTRACTION • REPLACE**

Final Standard & Verification

The roadmap succeeds only when learning is converted into secure, tested, documented and demonstrable professional capability.

10/10 roadmap definition: Anwar becomes a professional who can design, build, secure, automate, govern and improve AI-enabled cloud solutions; communicate technical and business trade-offs; and prove every major claim through Moltrix evidence.

Verification note

Credential plans are intentionally time-sensitive. Immediately before study activation or exam booking, verify names, objectives, prerequisites, availability and retirement notices on official Microsoft Learn, AWS Certification, HashiCorp Developer and CNCF/Linux Foundation pages. AB-650 remains provisional until official public credential materials are available and stable.

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